THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT MID TERM JOINT EXAMINATION

FORM ONE
033 BIOLOGY

Time: 2:30 Hours **Wednesday, 26th August 2026 p.m.**

Instructions

1. This paper consists of sections A, B, and C with a total of ten questions.
2. Answer **ALL** questions.
3. All answers must be written in the spaces provided.
4. Write your examination number on every page in this question paper.
5. All writing must be in blue/black ink. Diagrams must be drawn in pencil and labeled in ink.
6. Cell phones and calculators are not allowed in the examination room

QUESTION NUMBER	SCORE	INITIALS OF EXAMINER
1		
2		
3		
4		
5		
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7		
8		
9		
10		
TOTAL		

SECTION A (15 MARKS)

1. Choose the most correct answer from the alternatives given and write its letter in the answer table provided.

(i) During a community health campaign, students explain how knowledge of Biology can help people understand diseases, nutrition and proper hygiene. Which statement best describes Biology?
A. The study of rocks and minerals
B. The study of living organisms and their interactions
C. The study of weather and climate
D. The study of numbers and measurements

(ii) A student wants to investigate whether the amount of sunlight affects the growth of bean seedlings. Which of the following is the best first step in a scientific investigation?
A. Drawing a conclusion
B. Formulating a hypothesis
C. Recording the results
D. Writing the report

(iii) Seypalm wants to measure exactly **25 cm³** of water during a laboratory investigation. Which apparatus should he use?
A. Measuring cylinder
B. Thermometer
C. Balance
D. Hand lens

(iv) During a laboratory experiment, a student accidentally breaks a test tube. What should the student do first?
A. Pick up the broken pieces with bare hands
B. Hide the broken pieces
C. Inform the teacher or laboratory assistant
D. Continue with the experiment

(v) A student observes a very small organism using an instrument that magnifies objects and contains an objective lens and an eyepiece. Which apparatus is being used?
A. Hand lens
B. Microscope
C. Beam balance
D. Measuring cylinder

(vi) A student observes a cell that has a cell wall, chloroplasts and a large permanent vacuole. The cell is most likely a:
A. Red blood cell
B. Nerve cell
C. Plant cell
D. Sperm cell

(vii) A learner observes that cells in the root tip of a plant are actively dividing. Which characteristic of living things is demonstrated?
A. Respiration
B. Reproduction
C. Growth
D. Excretion

(viii) A student observes two organisms. One is made of one cell while the other consists of many cells with specialized functions. The process by which cells become specialized is called:
A. Diffusion
B. Cell differentiation
C. Photosynthesis
D. Germination

(ix) A farmer notices green plants growing well in an area receiving sufficient sunlight. Which process enables green plants to manufacture their own food?
A. Respiration
B. Photosynthesis
C. Excretion
D. Transpiration

(x) A student finds an organism with a microscopic body, no true nucleus and no membrane-bound organelles. To which kingdom does it most likely belong?
A. Plantae
B. Animalia
C. Monera
D. Fungi

Answer table

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)

2. Match each item in **List A** with the most appropriate response in **List B**. Write the letter of the correct response beside each item number.

LIST A	LIST B
(i) Organism made up of many cells	A. Binomial nomenclature
(ii) Scientific naming system using two names	B. Fungi
(iii) Organisms without a true nucleus	C. Multicellular organism
(iv) Organisms that obtain nutrients from dead organic matter	D. Monera
(v) Green organisms that manufacture their own food	E. Plantae
	F. Unicellular organism
	G. Animalia

Answer table

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B					

SECTION B (70 MARKS)

3. A Form One Biology class wants to investigate whether different amounts of water affect the growth of bean seedlings. The teacher provides bean seeds, containers, soil, water, a ruler and a balance.
(a) Identify **two** pieces of apparatus/resources from the scenario that can be used to obtain measurements during the investigation.
(i) ……..
(ii) ……..

(b) State **two** laboratory safety precautions the students should observe while carrying out the investigation.

(i) ….……

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(ii) ….……

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(c) Suggest a suitable **hypothesis** for the investigation.

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(d) Identify the following variables in the investigation:

(i) Independent variable

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(ii) Dependent variable

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(iii) One controlled variable

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(iv) One control group condition

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4. During a school career day, students discuss how Biology is useful in agriculture, medicine, environmental conservation and other areas of human life.

(a) Explain what is meant by Biology.

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(b) Give **four** ways in which studying Biology is important to an individual or society.

(i) ….……

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(ii) ….……

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(iii) ….……

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(iv) ….……

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(c) State **two scientific fields that are closely related to Biology** and explain one contribution of each to the study of living organisms.

(i) ….……

(ii) ……

5. A student observes two prepared specimens under a microscope. Specimen X has a cell wall, chloroplasts and a large vacuole, while specimen Y lacks these structures.

(a) Identify specimen X as either a plant cell or animal cell, giving **one reason** for your answer.

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(b) State four differences between plant cells and animal cells.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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(c) Explain the importance of cell differentiation in multicellular organisms. Give two points.

(i) ……

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(ii) ……

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6. A student collecting organisms around the school compound records the following observations:

(i) Organism P is microscopic and has no true nucleus.
(ii) Organism Q is a green organism that manufactures its own food.
(iii) Organism R obtains nutrients from dead organic matter.
(iv) Organism S is an organism made of many cells and is capable of movement.

(a) Name the kingdom to which each organism belongs:

(i) P:……

(ii) Q: ……

(iii) R: ……

(iv) S: ……

(b) Give three reasons why living organisms are classified into groups.

(i) ……

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(ii) ……

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(iii) ……

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(c) State the meaning of binomial nomenclature.

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(d) Give two advantages of using scientific names instead of common names.

(i) ……

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(ii) ……

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7. Seypalm health club is educating students about organisms that cause diseases. The students learn that among the infectious agents were viruses and bacteria.

(a) What is a virus?

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(b) State three characteristics of viruses.

(i) ……

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(ii) ……

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(iii) ……

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(c) Give two differences between viruses and bacteria.

(i) ……

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(ii) ……

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(d) State three major groups of living organisms studied in Biology, other than viruses.

(i) ……

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(ii) ……

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(iii) ……

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8. Kingdom Monera consists of primitive microscopic organisms such as bacteria.

(a) State three distinct characteristics of organisms belonging to Kingdom Monera.

(i) ……

(ii) ……

(iii) ……..

(b) Draw a well-labelled diagram of a typical bacterium cell.

(c) Explain two beneficial roles and two harmful roles of bacteria in daily human life or nature.

Beneficial roles

(i) ….……

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(ii) ….……

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Harmful roles

(i) ….……

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(ii) ….……

…….

(d) Name **one disease** caused by bacteria in humans.

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9. Kingdom Protoctista is another diverse group of living things studied under classification.

(a) State three characteristics of organisms belonging to Kingdom Protoctista.

(i) ….……

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(ii) ….……

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(iii) ….……

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(b) Apart from the kingdoms Monera, Protoctista, and Fungi, list the two remaining kingdoms of living things.

(i) ….……..

(ii) ……..

(c) Living things are grouped into hierarchical categories. Arrange the seven taxonomic ranks of classification in descending order, starting from the largest group to the smallest.

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SECTION C (15 MARKS)

10. An outbreak of a mysterious disease occurs in a local community. Patients show symptoms of high fever, and doctors discover that the infectious agent is microscopic, lacks cellular structures, does not respond to standard bacterial antibiotics, and can only replicate inside living host cells.

(a) Identify the group/type of infectious agent responsible for this outbreak.

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(b) Provide an argument supporting why this agent is considered non-living when outside a host cell, and another argument supporting why it shows characteristics of living things when inside a host cell.

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(c) Outline four general rules or guidelines that scientists must follow when writing scientific names of organisms using the Binomial Nomenclature system.

(i) ……

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(ii) ……

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(iii) ……

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(iv) ……

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(d) Explain three reasons why it is dangerous to handle unknown micro-organisms without proper laboratory safety equipment.

(i) ……

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(ii) ……

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(iii) ……

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